

DNA structure & variant explorer — final cheat sheet

Everything to hand your teammate. No biology background required to use this.

1. What each tool does (in plain words)

Ensembl = the library with the actual DNA letters (A/T/G/C) for any gene. Feeds the visual — this is what gets drawn as the helix.

MyVariant.info = the “what’s wrong here and why it matters” tool. Give it one specific known mutation ID, it returns whether it’s harmful, how common, what disease. Feeds the click-to-see-info popup.

2. Links and IDs — copy exactly as-is

Gene	Ensembl link (structure)	Demo variant	MyVariant.info link (meaning)
BRCA1	<code>rest.ensembl.org/sequence/id/ENSG00000012048?content-type=text/x-fasta</code>	rs80357906	<code>myvariant.info/v1/query?q=rs80357906</code>
HBB	Search “HBB” on ensembl.org to confirm gene ID, same URL pattern as BRCA1	rs334	<code>myvariant.info/v1/query?q=rs334</code>

Note: BRCA1 gene page on Ensembl shows location 17:43,044,292-43,170,245, gene ID ENSG00000012048. Look up HBB the same way on ensembl.org search.

3. Myotonic dystrophy — different rule, no API needed

This condition is caused by a repeating pattern (CTG) copied too many times, not a single changed letter or a deletion. There’s no single spot to click, so it doesn’t use either tool above. Just build a simple slider/bar with three colored zones based on the repeat count:

- Up to 37 repeats → normal (green)
- 38–49 repeats → pre-mutation, no symptoms yet, but risk for children (yellow)
- 50+ repeats → disease range, myotonic dystrophy (red)

4. Formula: where to highlight on the helix

The MyVariant.info response includes a field called **hg38** with a **start** position. This number counts from the start of the whole chromosome, not the gene. To find where inside the gene’s sequence to highlight, subtract the gene’s own start position:

`click position = variant's hg38 start position – gene's start position`

Example for the BRCA1 demo variant: 43,057,062 – 43,044,292 ≈ letter 12,770 in the downloaded sequence. Note: BRCA1 reads on the reverse strand, so this is an approximate guide, not pixel-perfect — that’s fine for a demo.

5. Exact words to hand your teammate

“Here’s a link that gives you a long string of the letters A, T, G, C — treat it as plain text. Draw one twisted-ladder rung per letter. Here’s a second link that returns a JSON object with a field called `clinical_significance` — when

someone clicks the rung near the calculated position, show that field in a popup. For the third gene (myotonic dystrophy), skip both links — just build a slider with three colored zones based on a repeat-count number I'll give you. No biology knowledge needed beyond this.”

Final reference sheet — Codex hackathon, DNA structure & variant explorer.